

PROJECT SEMESTER REPORT

PSPCL ERP Implementation

by

Puru Sachdeva

Roll No. 102103506

Under the Guidance of

Mr. Rahul Kumar Kushwah (Business Analyst)

Dr. Jyoti (Assistant Professor)



Submitted to the

**Computer Science & Engineering Department
Thapar Institute of Engineering & Technology, Patiala**

In Partial Fulfilment of the Requirements for the Degree of

Bachelor of Engineering in Computer Engineering

at

Thapar Institute of Engineering & Technology, Patiala

July 2025

PSPLC ERP Implementation

by Puru Sachdeva

Place of work: **AWC Software, PSPCL Head Office, Patiala**

Submitted to the Computer Science & Engineering Department, Thapar Institute of Engineering & Technology

June 2025

In Partial Fulfilment of the Requirements for the Degree of Bachelor of Engineering in Computer Engineering

Abstract:

This report outlines the work conducted during my internship at AWC Software, where I contributed to the ERP implementation project for Punjab State Power Corporation Limited (PSPCL). The primary focus of my internship was to design and maintain a centralized bug tracking system using MantisBT, aimed at improving the efficiency and reliability of ERP-related issue management.

My responsibilities included setting up the MantisBT environment, configuring user roles and access levels, creating project categories, and regularly updating and monitoring reported issues. The system enabled streamlined communication between development, testing, and support teams, replacing previously scattered issue-reporting methods. By logging detailed bug reports and tracking their resolution status, the team was able to significantly reduce resolution time and improve system stability.

The project offered valuable exposure to the practical application of software engineering concepts, particularly in issue tracking, user interaction, and ERP workflows. It also enhanced my skills in coordination, documentation, and tool configuration. Through this internship, I developed a better understanding of how large-scale government ERP systems operate and the importance of systematic defect management.

This report covers the project's background, methodology, key observations, limitations, and future scope for improvement in ERP support systems.

Author: Puru Sachdeva



Certified by: Mr. Rahul Kumar Kushwah
(Industrial Coordinator / Mentor)

Certified by: Dr. Jyoti
(Faculty Mentor)

CERTIFICATE (PROJECT SEMESTER TRAINING) FROM THE COMPANY OR THE ORGANIZATION



ORACLE Gold Partner
Technology - Outsourcing - Consulting - Implementation - Support

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Puru Sachdeva (Roll Number: 102103506)**, a student of the Computer Engineering department pursuing Bachelor of Engineering from Thapar Institute of Engineering and Technology, Patiala, has undertaken an internship as a Software Intern at AWC Software.

The internship began on **3rd February 2025** and is expected to conclude on **30th June 2025**, spanning a total duration of 5 months. **AWC Software Pvt. Ltd.** has no objection to the student submitting the project report to their institution, based on the work undertaken here.

We wish him continued success for the remaining duration of his internship and in all his future endeavors.

FOR AWC Software Pvt. Ltd.

Bhavneek Singh
(Human Resources)

AWC Software Pvt. Ltd.
ISO 9001 : 2008 Certified
CIN: U72900HR2009PTC039303

B-3, Sector 60, NOIDA, U.P. - 201301
Website : www.awcsoftware.com
Tel: 0120 - 4398300

TABLE OF CONTENT

1. Company Profile.....	1
2. Introduction	4
3. Background.....	6
4. Objectives	8
4.1. To support and streamline post-implementation issue tracking	8
4.2. To ensure process accuracy and system reliability	8
4.3. To improve inter-team coordination and operational accountability.....	8
5. Methodology.....	9
5.1. Post-implementation issue tracking using Mantis Bug Tracker	9
5.2. Functional validation of custom ERP forms (SR and SRW).....	10
5.3. Communication and coordination across ERP teams	10
6. Observations and Findings	13
6.1. Centralized issue tracking improved visibility and accountability	13
6.2. Recurring issues highlighted key pain points in the ERP implementation	13
6.3. User training and documentation had a direct impact on issue quality	14
6.4. MantisBT usage helped establish traceability and reporting standards	14
7. Limitations.....	15
7.1. Limited integration between MantisBT and Oracle EBS R12.....	15
7.2. Restricted access to backend systems	15
7.3. Basic reporting capabilities in MantisBT	15
7.4. Functional scope restricted to select modules and forms.....	16
8. Conclusions and Future Work	17
8.1. Key Conclusions	17
8.2. Future Work and Recommendations:	17
9. Bibliography/References	19

TABLE OF FIGURES

Figure 1 Oracle Partner	1
Figure 2 Microsoft Partner	2
Figure 3 Salesforce Partner	2
Figure 4 Oracle EBS Modules	4
Figure 5 Sample MantisBT Dashboard.....	7
Figure 6 Bug/Issue Life Cycle	11
Figure 7 User Hierarchy.....	12
Figure 8 Automation Workflow.....	18

1. Company Profile

AWC Software Pvt. Ltd. is a trusted IT services and consulting organization delivering enterprise-grade solutions to clients across public and private sectors. The company specializes in implementing, customizing, and supporting complex business systems that enable digital transformation, operational excellence, and data-driven decision-making.

With a strong presence in enterprise application development, cloud computing, CRM, and ERP solutions, AWC Software has built a reputation for delivering secure, scalable, and tailor-made solutions across domains such as power, finance, government, healthcare, and manufacturing.

AWC Software has established strategic partnerships with leading global technology providers:

- **Oracle Partner**

As an Oracle Partner, AWC holds deep expertise in Oracle technologies including **Oracle E-Business Suite (EBS R12)**, **Oracle Fusion**, **Oracle Cloud Infrastructure (OCI)**, and advanced **database management systems**. This partnership enables AWC to access technical resources, implementation tools, and certified training programs that support its delivery of end-to-end ERP, cloud, and data solutions. The company focuses on modules like Finance, Supply Chain Management (SCM), and Human Capital Management (HCM), offering tailored solutions to enterprise clients.



Figure 1 Oracle Partner

- **Microsoft Gold Partner**

AWC's collaboration with Microsoft demonstrates its capability to deliver cloud-native, intelligent business solutions using **Microsoft Azure**. Leveraging Microsoft's platforms and tools, the company helps clients scale infrastructure, streamline operations, and integrate AI and data analytics into their systems, contributing to productivity and cost efficiency.



Figure 2 Microsoft Partner

- **Salesforce Partner**

AWC is also a Salesforce partner, offering services around one of the world's leading **customer relationship management (CRM)** platforms. Through this alliance, the company assists clients in managing sales pipelines, marketing campaigns, customer service workflows, and end-user engagement using Salesforce's cloud-based ecosystem.



Figure 3 Salesforce Partner

Key Service Areas of AWC Software Include:

- **Enterprise Resource Planning (ERP)** – Implementation and support for Oracle EBS and Fusion across Finance, SCM/MM, and HRMS modules.
- **Database Management & Infrastructure** – Deployment and tuning of Oracle database solutions to ensure high performance, availability, and secure data handling.
- **Cloud Services** – Design and administration of scalable cloud environments using Oracle Cloud Infrastructure and Microsoft Azure.
- **CRM & Customer Experience** – Salesforce-based solutions for customer engagement, sales force automation, and campaign management.
- **Custom Development & Integration** – Development of tailor-made applications and seamless integration into existing enterprise ecosystems.

As part of its commitment to customer success, AWC Software provides both technical and functional support services, ensuring that client solutions are robust, adaptable, and aligned with long-term business goals.

In alignment with this vision, the internship project was associated with the implementation and stabilization of **Oracle EBS R12 for Punjab State Power Corporation Limited (PSPCL)**. The work primarily involved activities related to the **Materials Management (MM)** module and custom Oracle forms such as **Store Requisition (SR)** and **Store Return Warrant (SRW)**. The internship also included responsibility for managing and customizing the **Mantis Bug Tracker (MantisBT)** system used for ERP issue tracking — reflecting AWC’s emphasis on operational transparency, structured project support, and end-user satisfaction.

Through its strategic partnerships and technological depth, AWC Software continues to empower businesses in their journey toward digital innovation and enterprise modernization.

2. Introduction

Enterprise Resource Planning (ERP) systems have become essential for large-scale organizations to efficiently manage business functions across departments through a centralized platform. Their relevance is especially pronounced in public sector utilities, where diverse operations like procurement, finance, inventory management, and human resources must operate cohesively under strict regulatory and service-level requirements.

Punjab State Power Corporation Limited (PSPCL), the state-run electricity utility for Punjab, initiated a major digital transformation by implementing **Oracle E-Business Suite Release 12 (EBS R12)**. This ERP system integrates a wide range of business modules to streamline operations, eliminate redundancy, and improve transparency. The implementation was carried out with the support of **AWC Software Pvt. Ltd.**, a trusted **Oracle Gold Partner** with deep expertise in ERP customization and support.

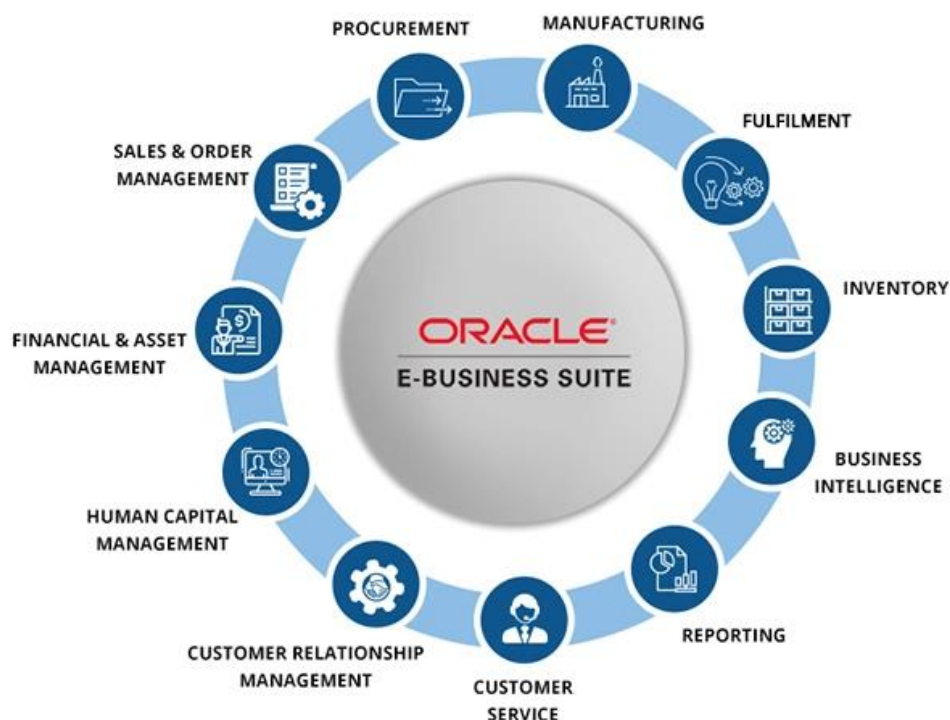


Figure 4 Oracle EBS Modules

The internship project was embedded within this ERP transformation initiative and focused on several real-world challenges encountered during implementation and support activities. It offered hands-on exposure to the **technical infrastructure**, **functional workflows**, and

collaborative processes needed to successfully run and stabilize an ERP system in a public sector environment.

The key areas covered during the internship included:

- **Deployment and Configuration of Mantis Bug Tracker (MantisBT):**

Implemented and customized MantisBT to serve as a centralized system for reporting, tracking, and managing software bugs across various Oracle EBS modules.

- **Defect Management and Workflow Coordination:**

Participated in the full lifecycle of defect tracking — from issue logging and categorization to assignment, resolution follow-ups, and closure documentation.

- **Form Validation and Functional Workflow Testing:**

Contributed to the testing and validation of two important inventory-related forms:

- **Store Requisition (SR) Form** – for internal material requests
- **Store Return Warrant (SRW) Form** – for returning surplus or faulty stock

- **User Onboarding and System Documentation:**

Assisted in preparing user guides and training materials to help technical and functional users navigate the bug tracking system effectively.

This internship served as a bridge between academic learning and real-world enterprise systems. It provided direct exposure to:

- Live ERP modules in a high-stakes environment
- Interdepartmental collaboration in public sector IT projects
- Issue tracking as a critical support mechanism in ERP rollouts
- Form-based process mapping within inventory and supply chain workflows

Overall, the experience reflected the multi-faceted nature of ERP implementation, where technology, process, people, and continuous feedback loops come together to drive organizational efficiency.

3. Background

Public sector enterprises increasingly rely on integrated ERP systems to ensure operational efficiency, process transparency, and real-time data-driven decisions. **Punjab State Power Corporation Limited (PSPCL)**, a state-owned electricity utility, initiated a significant digital transformation initiative by implementing **Oracle E-Business Suite Release 12 (EBS R12)** to centralize its core business processes.

[1]

The ERP rollout at PSPCL focused on key modules:

- **Human Resource Management System (HRMS)** – to manage employee records, payroll, attendance, and recruitment.
- **Finance** – to streamline accounting, budgeting, and financial reporting processes.
- **Materials Management (MM)** – to digitize inventory control, purchasing, and internal store workflows.

The transition from fragmented legacy systems to an integrated Oracle ERP platform was a complex and multi-phased effort. Like most enterprise-scale implementations, the project faced several technical and functional challenges during and after rollout, including data inconsistencies, workflow gaps, and system misconfigurations.

To manage these issues in a structured, transparent, and accountable way, the organization adopted **Mantis Bug Tracker (MantisBT)** as its centralized issue tracking solution.

MantisBT is an open-source, web-based issue tracking system widely used in both enterprise and development environments. It enables organizations to:

- Log software bugs, feature requests, and support issues.
- Assign tasks to responsible teams or individuals with deadline tracking.
- Monitor progress through well-defined status transitions (e.g., new → acknowledged → resolved → closed).
- Generate summary and analytics reports on open, resolved, and overdue issues.

In the context of ERP support at PSPCL, MantisBT served as a critical tool for:

- Documenting issues related to Oracle EBS R12 modules (Finance, MM, HRMS).
- Categorizing problems by severity, module, and user group.
- Facilitating inter-team collaboration between functional and technical units.
- Enhancing visibility and control over the post-implementation support lifecycle.

With MantisBT in place, the ERP team could ensure that all reported issues were centrally captured, assigned, and resolved in a timely and auditable manner — a key factor in stabilizing the live system and building trust among end-users.

Additionally, MM-specific workflows such as **Store Requisition (SR)** and **Store Return Warrant (SRW)** forms often generated high volumes of transactional data and occasional workflow exceptions. MantisBT helped track and resolve such module-specific issues efficiently, improving user adoption and reducing downtime.

The inclusion of a formal bug tracking system like MantisBT reflects industry best practices in ERP deployment and highlights the importance of operational governance, especially in mission-critical public infrastructure projects.

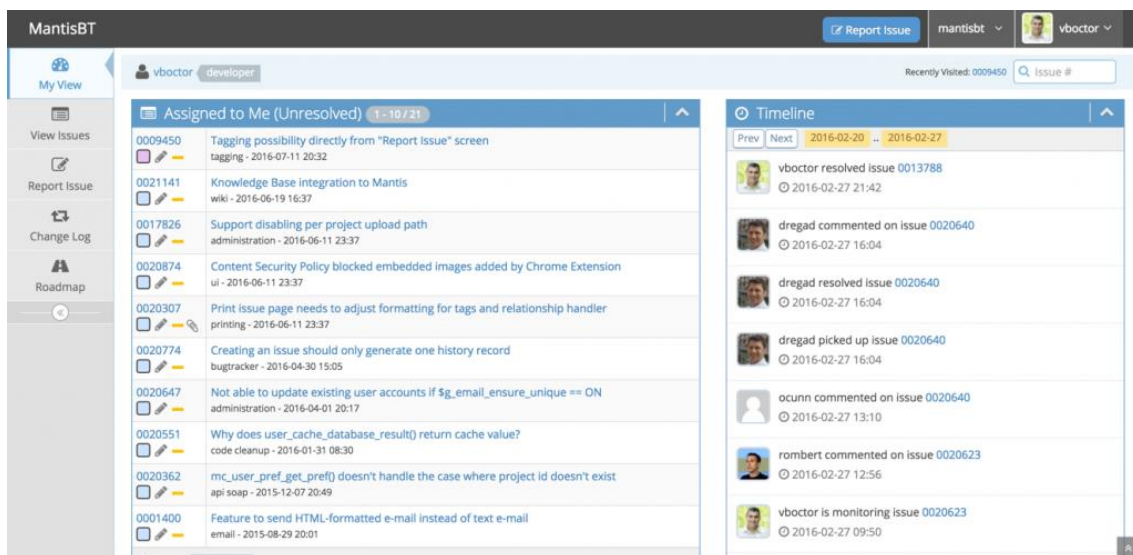


Figure 5 Sample MantisBT Dashboard

4. Objectives

The primary objectives of this project semester were centred around supporting the successful implementation and stabilization of ERP systems at PSPCL using Oracle E-Business Suite R12. The specific objectives were:

4.1.To support and streamline post-implementation issue tracking

- Configure and customize Mantis Bug Tracker (MantisBT) to align with PSPCL's module-wise issue categorization.
- Enable role-based access and workflows for issue lifecycle management (e.g., open → acknowledged → resolved → closed).
- Improve visibility, prioritization, and resolution tracking for ERP-related technical and functional issues

4.2.To ensure process accuracy and system reliability

- Assist in testing and verifying the behaviour of custom forms such as the Store Requisition (SR) and Store Return Warrant (SRW) forms.
- Identify mismatches or gaps in expected form functionality versus actual system behaviour.
- Report identified issues through MantisBT for appropriate resolution and follow-up.

4.3.To improve inter-team coordination and operational accountability

- Maintain accurate and up-to-date records of logged issues and their resolution statuses.
- Serve as a link between technical developers and functional teams by ensuring communication around bug resolution.
- Enhance the reliability and stability of live Oracle ERP modules through continuous feedback and support.

5. Methodology

The methodology followed during the internship was structured around achieving three key goals: supporting ERP issue tracking, validating business workflows, and improving communication between functional and technical teams. The approach was practical, iterative, and closely aligned with real-time ERP operations at PSPCL.

5.1. Post-implementation issue tracking using Mantis Bug Tracker

To manage issues effectively across Oracle EBS R12 modules, MantisBT was deployed and customized. The steps included:

- Installing and configuring MantisBT on the internal environment with module-wise project categories (e.g., HRMS, Finance, MM).
- Regularly logging issues raised by stakeholders, categorizing them by severity and department, and assigning them to relevant teams.
- Monitoring issue resolution, updating statuses, and generating summary reports to track performance over time.
- Mapping issue status flows (e.g., new → acknowledged → confirmed → assigned → resolved → closed) to support the typical lifecycle of ERP support tickets.
- Defining user roles such as reporter, developer, tester, and manager, along with corresponding permissions.

5.2.Functional validation of custom ERP forms (SR and SRW)

As part of the MM module workflows, two key inventory-related forms were evaluated for correctness and usability:

- Tested the Store Requisition (SR) form to verify proper flow of material request initiation and approval.
- Reviewed the Store Return Warrant (SRW) form to ensure that return processes were correctly aligned with inventory rules and business logic.
- Reported any bugs or process mismatches identified during form usage through MantisBT for resolution by the technical team.
- Collaborated with functional consultants to understand form validation requirements and assist in ensuring compliance with PSPCL's business processes.

5.3.Communication and coordination across ERP teams

Maintaining smooth communication between technical developers, functional users, and management was essential for timely issue resolution:

- Maintained a consistent record of reported and resolved issues with clear categorization and comments for historical reference.
- Participated in team discussions to clarify issue details, replicate bugs, and suggest improvements where appropriate.
- Followed up on overdue or high-priority issues to ensure timely escalation and closure, improving trust and system adoption among users.
- Shared feedback from end-users with developers to bridge the communication gap between business needs and technical implementation.

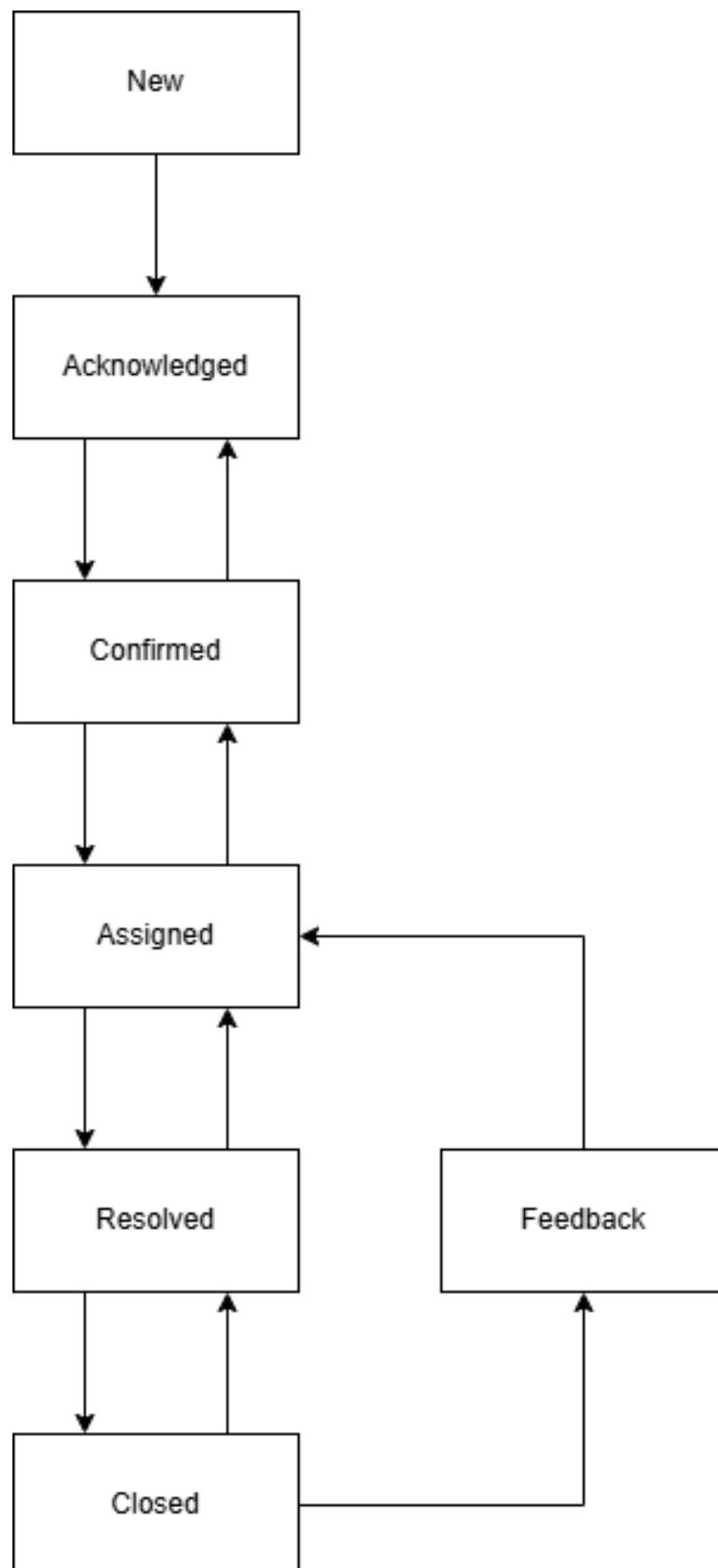


Figure 6 Bug/Issue Life Cycle

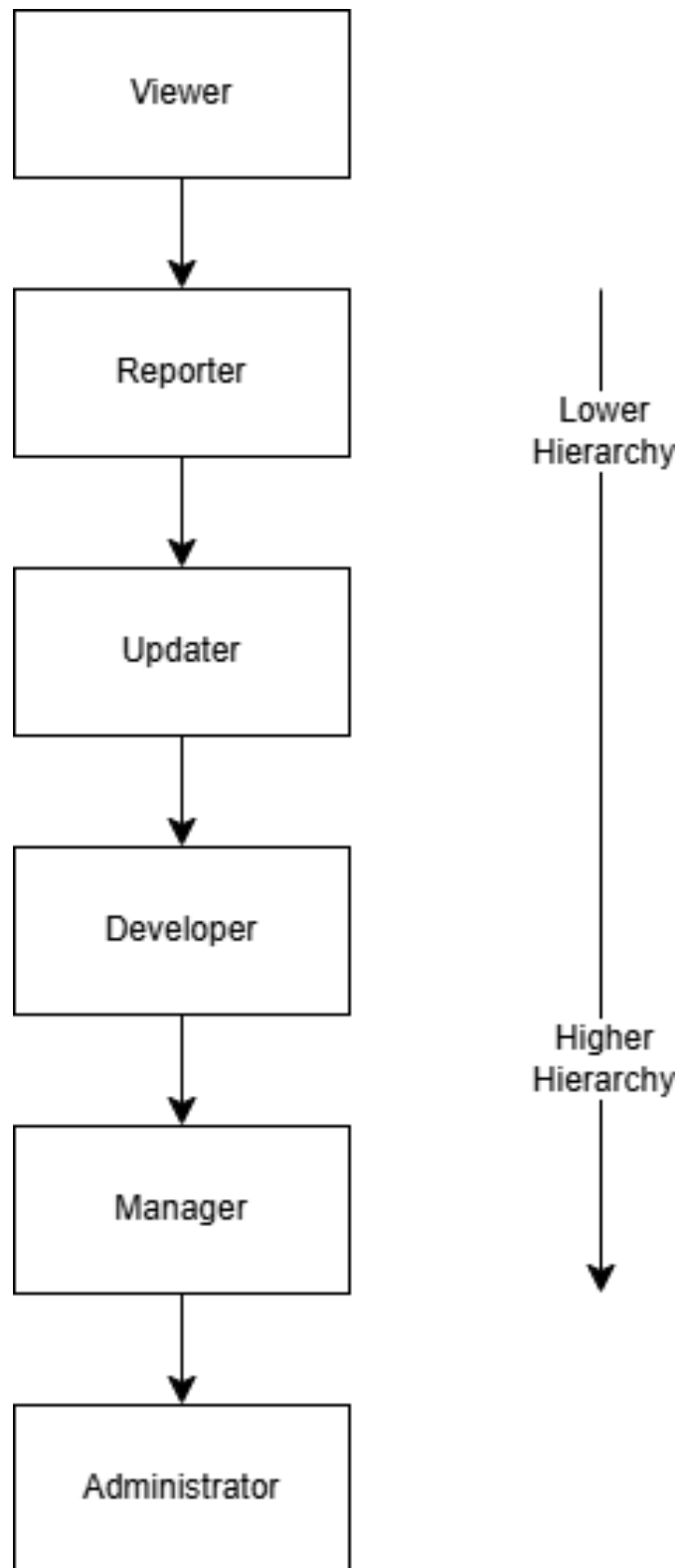


Figure 7 User Hierarchy

6. Observations and Findings

During the internship, several important insights were gathered across the areas of ERP issue tracking, functional workflow validation, and team collaboration. These observations reflect both system-level patterns and user behaviour, which are critical to improving ERP stability and long-term adoption.

6.1. Centralized issue tracking improved visibility and accountability

- The implementation of MantisBT helped establish a structured, single-point system for tracking and managing defects in Oracle EBS R12 modules.
- Functional and technical teams were able to view, assign, and monitor issues more effectively, reducing reliance on scattered communication via email or verbal instructions.
- The categorization of issues by module (Finance, HRMS, MM) and severity improved prioritization and resource planning.

6.2. Recurring issues highlighted key pain points in the ERP implementation

- A significant number of issues were related to data migration mismatches, especially in Finance and Inventory modules.
- User-reported bugs often stemmed from misconfigured approval hierarchies, incorrect report generation, and missing validations in custom forms.
- The Store Requisition (SR) and Store Return Warrant (SRW) forms had a few common issues such as missing mandatory field checks, improper stock balances, and incorrect document status transitions.

6.3. User training and documentation had a direct impact on issue quality

- Users who received demonstrations or reference guides submitted more detailed and well-structured bug reports, leading to faster resolution.
- Lack of awareness in some departments led to duplicate entries or vague problem descriptions, which delayed issue assignment and closure.
- Training improved the adoption of the MantisBT system, making it easier to gather reliable defect data across modules.

6.4. MantisBT usage helped establish traceability and reporting standards

- Audit trails and comment logs within MantisBT made it easy to trace the communication history and resolution path for each issue.
- Summary reports generated from MantisBT helped project leads assess team responsiveness and identify bottlenecks.
- The visibility of unresolved issues served as a real-time indicator of areas where additional testing or process refinement was required.

The exact findings cannot be shown due to AWC's confidentiality policies. However, the findings shown highlight the value of structured issue management tools in large-scale ERP projects. They also emphasize the importance of user engagement, clear workflows, and collaborative communication between technical and functional teams.

7. Limitations

While the internship contributed to meaningful improvements in ERP issue tracking and functional validation, certain limitations were encountered that constrained the overall scope and efficiency of the work:

7.1. Limited integration between MantisBT and Oracle EBS R12

- MantisBT operated as a standalone platform and did not directly integrate with Oracle ERP modules.
- All issues had to be manually logged by users or support staff, which introduced the risk of data entry errors and occasional duplication.
- Lack of system-level alerts from Oracle meant that issues discovered during ERP usage were often reactive rather than proactively flagged.

7.2. Restricted access to backend systems

- As part of standard security policy, interns were not granted direct access to the Oracle database, application server, or system-level configuration files.
- This limited the ability to perform deeper analysis or debug certain technical issues independently, requiring dependencies on senior technical staff.

7.3. Basic reporting capabilities in MantisBT

- MantisBT's default reporting features offered limited visual analytics, such as charts or KPI dashboards.
- Generating advanced summaries for management review required exporting data manually and creating custom reports using spreadsheet tools.
- Real-time progress dashboards were unavailable, which made tracking large issue queues slightly cumbersome.

7.4.Functional scope restricted to select modules and forms

- The internship work focused primarily on Finance, HRMS, and MM modules, particularly the SR and SRW forms.
- Other modules within Oracle EBS R12 (e.g., Projects, Procurement, Payroll) were outside the assigned scope and thus not explored in depth.
- Similarly, exposure to testing automation, performance tuning, or end-user training delivery was limited during the internship period.

These limitations were mostly structural and context-driven, but they provided valuable insights into the real-world challenges of working within enterprise IT environments — particularly in high-security, large-scale government projects.

8. Conclusions and Future Work

The internship at AWC Software, aligned with the Oracle EBS R12 implementation for Punjab State Power Corporation Limited (PSPCL), provided valuable exposure to enterprise-level software systems, structured issue management, and real-world IT operations. The primary areas of engagement—bug tracking through MantisBT, validation of inventory workflows (SR and SRW forms), and coordination between functional and technical teams—offered a strong foundation in both technical and process-oriented aspects of ERP support.

8.1.Key Conclusions

- **Structured issue tracking systems are essential** for stabilizing large ERP implementations. MantisBT played a pivotal role in improving visibility, accountability, and timely resolution of defects across Finance, HRMS, and MM modules.
- **Module-specific testing and validation**—especially of critical business forms like Store Requisition (SR) and Store Return Warrant (SRW)—helped identify operational gaps early and contributed to smoother process execution.
- **Cross-functional collaboration** was a key driver of success. Effective communication between developers, functional consultants, and support staff led to faster turnaround on reported issues and helped in building confidence among end-users.
- **Hands-on use of real systems** and tools deepened the understanding of ERP architecture, support workflows, and IT governance in a large public-sector environment.

8.2.Future Work and Recommendations:

While the internship achieved its core objectives, several opportunities for improvement and future enhancement were identified:

- **Integration between MantisBT and Oracle EBS R12**

A future improvement could involve semi-automated logging of ERP errors or system logs directly into MantisBT using middleware scripts or APIs. This would reduce manual entry and improve response time.

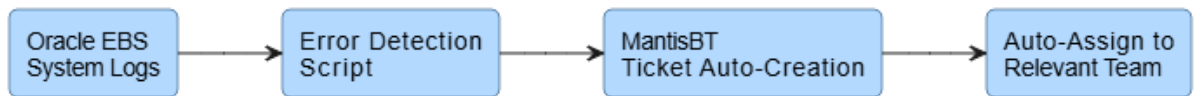


Figure 8 Automation Workflow

- **Advanced reporting and dashboards**

Integration with tools like Power BI or Excel-based macros can be used to create real-time dashboards and module-wise defect heatmaps to help management make informed decisions.

- **Enhanced training and onboarding materials**

Creating short video tutorials or in-system tooltips for MantisBT users could improve adoption and reporting quality, especially for non-technical stakeholders.

- **Extending coverage to other modules**

Future internships or support cycles could include deeper involvement in additional Oracle modules like Procurement, Payroll, or Project Costing to gain a more holistic view of the ERP ecosystem.

Overall, the internship served as a strong stepping stone into enterprise software practices, issue lifecycle management, and real-world collaboration. It demonstrated the importance of combining technical skill with structured processes to ensure the successful stabilization and support of large-scale ERP systems.

9. Bibliography/References

- [1] “Official Website of PSPCL,” 2025. [Online]. Available: <https://pspcl.in/>.
- [2] “Partners - AWC Software,” 2025. [Online]. Available: <https://awcsoftware.com/partners/>.
- [3] “AWC Software,” 2025. [Online]. Available: <https://awcsoftware.com/>.
- [4] “MantisBT UI,” 2025. [Online]. Available: <https://mantisbt.org/blog/archives/mantisbt/443>.
- [5] “Oracle Help Center – ERP Cloud and EBS R12 Product Guides,” Oracle, 2025. [Online]. Available: <https://docs.oracle.com>.
- [6] “MantisBT Official Documentation,” MantisBT, 2025. [Online]. Available: <https://www.mantisbt.org/documentation.php>.
- [7] “ERP Basics and Modules Overview,” TutorialsPoint, 2025. [Online]. Available: <https://www.tutorialspoint.com/erp/index.htm>.

Annexure A. Evaluation Form for Peer Review

Name of the student: (to be reviewed)	Puru Sachdeva	Roll no. of the student:	102103506
<i>This form has to be submitted by the student whose roll no. will be mentioned in the box above. Handover this to the panel at the time of final presentation.</i>			
Title of the project:	PSPCL ERP Implementation		
Name of the company:	AWC Software Pvt. Ltd.		
Project report (Tick the appropriate)	Excellent ✓	Good	Average
Project poster (Tick the appropriate)	Excellent ✓	Good	Average
Project video (Tick the appropriate)	Excellent ✓	Good	Average
Rate the work done	0 – 10 points	(Provide rating here) →	10
Give marks to the student on the basis of the overall performance	0 -5 marks	(Provide marks here) →	5
Abstract of the project (max. 100 words): The project aimed to support and enhance an ERP system developed for PSPCL. A major focus was on implementing effective issue management through Mantis Bug Tracker, enabling better tracking, assignment, and resolution of software bugs. The project also involved working with Service Requests and maintaining Oracle Reports (SRW Forms) to support operational and reporting needs within the ERP system.			
Mention three strengths of the work done: - Efficient Bug Tracking: Used Mantis Bug Tracker to log and manage issues systematically. - Improved Workflow: Helped streamline Service Request and report handling. - Better Collaboration: Enabled clear team communication and issue traceability.			
Provide some useful recommendations (It may be some improvements, some suggestions to further raise the quality of the project): To enhance the project further, integrating automated bug reporting with Mantis can reduce manual effort. Upgrading from Oracle Reports to modern BI tools would provide more user-friendly and insightful reporting. Additionally, extending full support for macOS would improve cross-platform compatibility and accessibility for a wider range of users.			
Name of the evaluator student:	Vashuki Achari	Roll no. of the evaluator student:	102103503
Signature of the Evaluator student:			